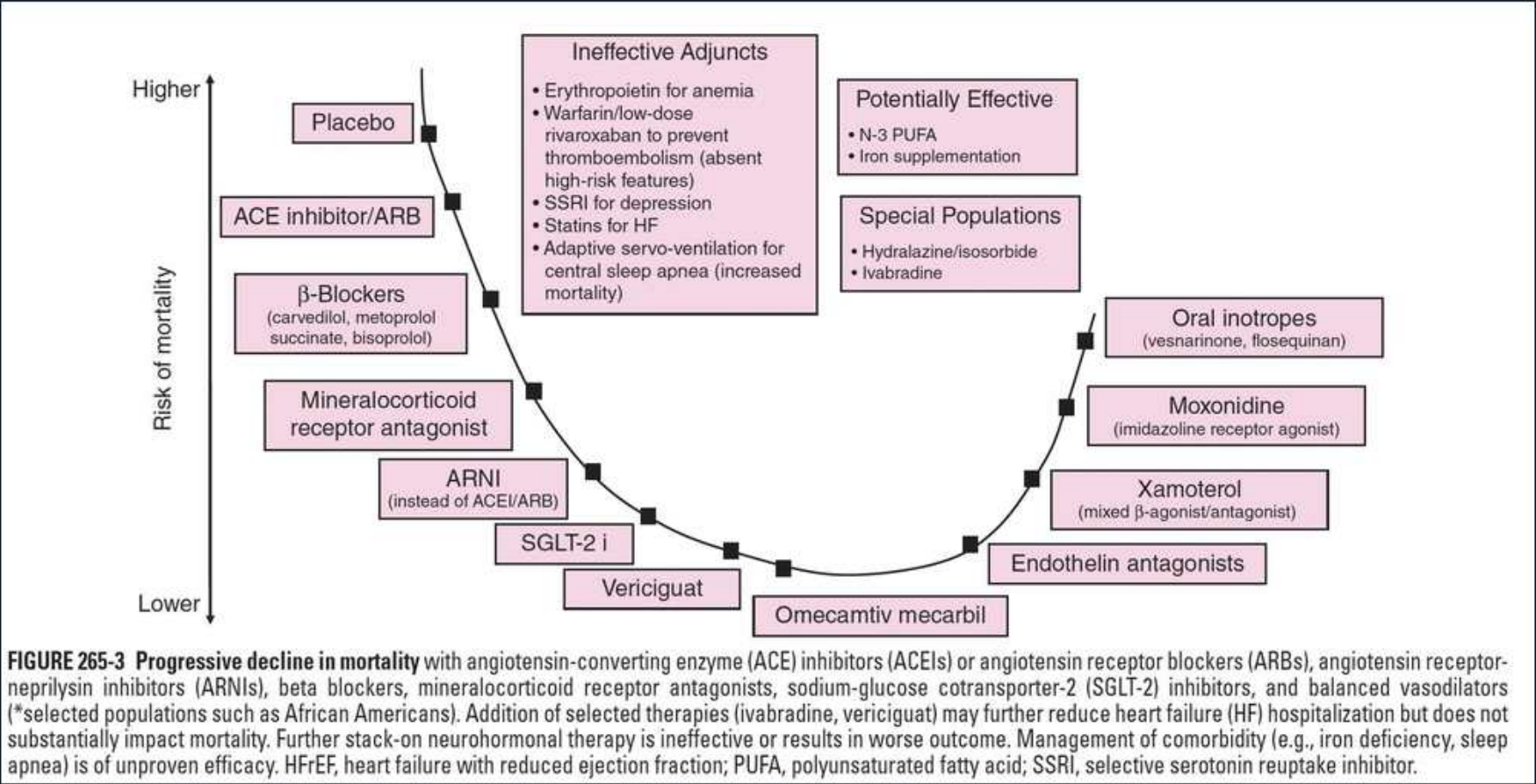


Heart Failure — Figure 265-3: Progressive Decline in HFrEF Mortality



1. Placebo — highest risk

WHAT YOU SEE

A pink 'Placebo' box pinned at the very TOP of the U-shaped curve, beside the 'Higher' end of the Risk-of-mortality arrow — its height marks it as the worst-survival reference point.

WHAT IT ENCODES

Placebo anchors the top of the U-shaped curve as the highest-mortality reference: untreated symptomatic HFrEF carries roughly 50% 5-year mortality, split between sudden cardiac death and progressive pump failure. Each proven drug class steps the patient stepwise DOWN the descending limb toward lower risk; harmful agents push UP the ascending limb. The figure's whole point is the additive, incremental nature of mortality reduction.

BOARD TRAP

WRONG answer: 'a single agent (e.g., a diuretic) is enough to substantially lower HFrEF mortality.' Discriminator: mortality reduction is cumulative across the four proven pillars; diuretics and most single non-pillar agents do not move the patient off the high-mortality placebo reference.

2. ACE inhibitor / ARB

WHAT YOU SEE

A pink 'ACE inhibitor/ARB' box one step DOWN the descending limb from placebo — dropping lower on the curve = the first mortality reduction.

WHAT IT ENCODES

ACE inhibitors (enalapril, lisinopril) are the first mortality-reducing step (CONSENSUS, SOLVD), blocking angiotensin II–driven vasoconstriction and adverse remodeling; ARBs (candesartan, valsartan) substitute when ACEi causes cough or angioedema. This rung is the historic RAAS-blockade foundation, now upgraded to ARNI when tolerated.

BOARD TRAP

WRONG answer: 'an ARB provides greater mortality benefit than an ACE inhibitor and should be first-line.' Discriminator: ARBs are non-inferior alternatives reserved for ACEi intolerance, not superior; the superior RAAS agent is ARNI, not an ARB.

3. Beta-blockers

WHAT YOU SEE

A pink 'β-Blockers (carvedilol, metoprolol succinate, bisoprolol)' box a further step DOWN the curve — lower position = added survival benefit stacked on RAAS blockade.

WHAT IT ENCODES

Only carvedilol, metoprolol SUCCINATE, and bisoprolol have proven mortality reduction in HFrEF (COPERNICUS, MERIT-HF, CIBIS-II) by countering chronic adrenergic toxicity and reversing remodeling. Initiate only when euvolemic and clinically stable, start low and titrate slowly; do NOT start during acute decompensation.

BOARD TRAP

WRONG answer: 'metoprolol tartrate or atenolol is acceptable for HFrEF mortality benefit.' Discriminator: only the three evidence-based agents (carvedilol, metoprolol succinate, bisoprolol) qualify; other β-blockers lack proven survival benefit and are class-effect traps.

4. MRA

WHAT YOU SEE

A pink 'Mineralocorticoid receptor antagonist' box continuing DOWN the descending limb — each lower rung is one more increment of mortality reduction.

WHAT IT ENCODES

Mineralocorticoid receptor antagonists — spironolactone (RALES) and eplerenone (EMPHASIS-HF/EPHESUS) — add further mortality reduction by blocking aldosterone-mediated fibrosis and remodeling in NYHA II–IV with EF ≤35%. Require K+ <5.0 and eGFR >30 with ongoing monitoring; eplerenone avoids the gynecomastia seen with spironolactone.

BOARD TRAP

WRONG answer: 'add an MRA without checking potassium or renal function.' Discriminator: MRAs are contraindicated with K+ ≥5.0 or eGFR <30 because of hyperkalemia risk; baseline and follow-up K+/creatinine checks are the gatekeeping step.

5. ARNI instead of ACEi/ARB

WHAT YOU SEE

A pink 'ARNI (instead of ACEi/ARB)' box low on the descending limb — its label literally says it REPLACES the ACEi/ARB rung, sitting lower (better survival) than it.

WHAT IT ENCODES

ARNI (sacubitril/valsartan) REPLACES the ACEi/ARB rung — PARADIGM-HF showed ~20% lower CV death/HF hospitalization versus enalapril. Neprilysin inhibition augments natriuretic peptides (vasodilation, natriuresis). Requires a ≥36-hour washout when switching FROM an ACEi to avoid additive angioedema; BNP becomes uninterpretable on ARNI (use NT-proBNP).

BOARD TRAP

WRONG answer: 'ARNI is added on top of an ACE inhibitor for combined RAAS blockade.' Discriminator: ARNI substitutes for, never combines with, an ACEi — concurrent use causes serious angioedema risk and requires a 36-hour ACEi washout before starting.

6. SGLT2i

WHAT YOU SEE

A pink 'SGLT-2 i' box near the BOTTOM of the curve — the deep position marks the 4th pillar at the lowest-mortality zone.

WHAT IT ENCODES

SGLT2 inhibitors (dapagliflozin DAPA-HF, empagliflozin EMPEROR-Reduced, 10 mg daily) form the 4th pillar, reducing CV death and HF hospitalization REGARDLESS of diabetes status via natriuresis and improved myocardial energetics. Adverse effects: genital mycotic infections, volume depletion, and rare euglycemic DKA (hold during acute illness/surgery).

BOARD TRAP

WRONG answer: 'reserve SGLT2 inhibitors for HFrEF patients who also have type 2 diabetes.' Discriminator: the trials proved benefit independent of diabetes, so non-diabetic HFrEF patients should still receive an SGLT2i — restricting to diabetics is the trap.

7. Vericiguat

WHAT YOU SEE

A pink 'Vericiguat' box sitting at the lowest part of the curve trough — bottom position = best-survival zone, an add-on after the four pillars.

WHAT IT ENCODES

Vericiguat is an oral soluble guanylate cyclase (sGC) stimulator that enhances the NO–cGMP pathway. In VICTORIA (high-risk worsening HFrEF) it reduced the composite of CV death or HF hospitalization, driven mainly by FEWER hospitalizations, but did NOT substantially reduce all-cause mortality. It is an add-on for recently decompensated patients already on the four pillars.

BOARD TRAP

WRONG answer: 'vericiguat strongly lowers mortality in HFrEF.' Discriminator: its benefit is largely on HF hospitalization/composite endpoint, not a substantial mortality reduction — don't choose it over a missing core pillar to save lives.

8. Omecamtiv mecarbil

WHAT YOU SEE

A pink 'Omecamtiv mecarbil' box at the very BOTTOM of the U (the trough) — lowest point of the curve, the floor of the descending limb.

WHAT IT ENCODES

Omecamtiv mecarbil is a selective cardiac myosin activator that prolongs systolic ejection and improves contractility WITHOUT raising intracellular calcium or myocardial oxygen demand (unlike classic inotropes). GALACTIC-HF showed a modest reduction in the HF-hospitalization/CV-death composite but no substantial all-cause mortality benefit. It is an investigational/niche add-on, not a pillar.

BOARD TRAP

WRONG answer: 'omecamtiv mecarbil is a safe oral inotrope that improves survival, unlike vesnarinone.' Discriminator: although mechanistically safer than calcium-raising inotropes, omecamtiv did not substantially reduce mortality — it is not a survival-improving drug and not a substitute for GDMT.

9. Ineffective adjuncts

WHAT YOU SEE

Pink box: EPO, warfarin/rivaroxaban, SSRI, statins, adaptive servo-ventilation

WHAT IT ENCODES

Ineffective (or harmful) in HFrEF: erythropoietin for anemia (RED-HF — no benefit, more thromboembolism), low-dose anticoagulation/warfarin or rivaroxaban without a high-risk feature (WARCEF/COMMANDER-HF — no net benefit), SSRIs for depression (no outcome benefit), statins for HF itself (CORONA/GISSI-HF — neutral; still treat coexisting ASCVD), and adaptive servo-ventilation, which actually INCREASES mortality in HFrEF with central sleep apnea (SERVE-HF).

BOARD TRAP

WRONG answer: 'prescribe adaptive servo-ventilation for central sleep apnea in this HFrEF patient,' or 'start a statin to improve HF survival.' Discriminator: ASV is HARMFUL (raises mortality) in HFrEF with central sleep apnea, and statins are neutral for HF itself (used only for ASCVD) — both are classic 'looks helpful but isn't' traps.

10. Potentially effective

WHAT YOU SEE

Box: N-3 PUFA, iron supplementation

WHAT IT ENCODES

Potentially/modestly effective: N-3 PUFA (omega-3 fish oil, GISSI-HF) gives a small reduction in death/hospitalization. IV iron — ferric carboxymaltose (FAIR-HF, AFFIRM-AHF) — repletes iron-deficient HFrEF patients (ferritin <100, or 100–300 with TSAT <20%) to improve symptoms, functional capacity, and reduce hospitalizations, EVEN without anemia.

BOARD TRAP

WRONG answer: 'give oral iron to the iron-deficient HFrEF patient.' Discriminator: oral iron (IRONOUT-HF) failed to replete stores or help outcomes in HF; the evidence-based route is IV iron, and repletion is indicated even with a normal hemoglobin.

11. Special populations

WHAT YOU SEE

Box: hydralazine/isosorbide, ivabradine

WHAT IT ENCODES

Special-population therapies. Hydralazine + isosorbide dinitrate (A-HeFT) reduces mortality especially in self-identified African-American patients already on GDMT, and serves as the vasodilator alternative when ACEi/ARB/ARNI are contraindicated (advanced renal failure, hyperkalemia, angioedema). Ivabradine (SHIFT) is added when the patient is in SINUS rhythm with resting HR ≥70 despite maximally tolerated β-blockade — it slows the SA node via the If current with no inotropic effect.

BOARD TRAP

WRONG answer: 'add ivabradine to control rate in HFrEF with atrial fibrillation.' Discriminator: ivabradine acts on the SA node and works ONLY in sinus rhythm with HR ≥70 on max β-blocker; it is ineffective for rate control in atrial fibrillation.

12. Oral inotropes — harmful

WHAT YOU SEE

Right side rising arm: vesnarinone, flosequinan

WHAT IT ENCODES

On the ascending (harm) limb: chronic oral inotropes — vesnarinone and flosequinan (and long-term oral milrinone, PROMISE) — INCREASE mortality through proarrhythmia and raised myocardial oxygen demand. IV inotropes (dobutamine, milrinone) are restricted to short-term support in cardiogenic shock/low-output states or palliative bridging, never chronic ambulatory therapy.

BOARD TRAP

WRONG answer: 'a chronic oral inotrope will support this stable HFrEF patient's contractility and prognosis.' Discriminator: long-term oral inotropes raise mortality; their only valid uses are short-term IV in shock or palliation/bridge-to-transplant — not chronic oral outpatient use.

13. Moxonidine / xamoterol — harmful

WHAT YOU SEE

Moxonidine, xamoterol on harm limb

WHAT IT ENCODES

Also on the harm limb: moxonidine (a central imidazoline/sympatholytic agonist) increased mortality in MOXCON, and xamoterol (a partial β -agonist) increased mortality as well. They demonstrate that misdirected neurohormonal modulation — central sympatholysis or partial β -agonism — is dangerous, in stark contrast to the proven benefit of full β -blockade.

BOARD TRAP

WRONG answer: 'a partial β -agonist like xamoterol offers gentle inotropic support with less risk than full agonists.' Discriminator: xamoterol and moxonidine both increased mortality in HFrEF; partial β -agonism is harmful, the opposite of evidence-based full β -blockade.

14. Endothelin antagonists — harmful

WHAT YOU SEE

A pink 'Endothelin antagonists' box on the RIGHT, ASCENDING (harm) limb of the U — climbing back up the curve = rising mortality.

WHAT IT ENCODES

Endothelin receptor antagonists (bosentan, darusentan) worsen HFrEF outcomes — despite endothelin's role as a potent vasoconstrictor, blocking it caused fluid retention and worsening heart failure in trials, with no survival benefit. They have no role in left-heart failure and belong instead to pulmonary arterial hypertension therapy.

BOARD TRAP

WRONG answer: 'block endothelin with bosentan to counter vasoconstriction in HFrEF.' Discriminator: endothelin antagonists cause fluid retention and worse HF outcomes; their evidence-based indication is PAH, not HFrEF.